

Hector E. Delgado Diaz

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Education

- Ph.D. University of Washington**, Astronomy and Astrobiology Sept 2019 – present
- M.S. in Astronomy degree conferred 2021-03
 - GPA: 3.71/4.00
- M.S. California State University Los Angeles**, Physics Aug 2017 – May 2019
- Dissertation Title: Stability of Nitrogen in Planetary Atmospheres in Contact with Liquid Water
- B.Sc. Universidad de Puerto Rico en Cayey**, General Sciences Aug 2012 – May 2017

Academic Experience

- University of Washington**, Research Assistant Seattle, WA, USA
Sept 2019 – present
- VPLanet Group; Advisor: Rory Barnes
 - Using VPLanet, an open source code written in C, developed simulations of exoplanets orbiting M dwarf stars and studied their habitability based on stellar, orbital, rotational, and atmospheric parameters.
 - With Python, created simulations of exoplanetary atmospheric spectra and used bayesian approach to simulate JWST observations of such atmosphere to assess habitability.
- University of Washington**, Instructor Seattle, WA, USA
Sept 2024 – present
- Co-Instructor of Cosmology (ASTR 425) with Dr. Jessica Werk during Winter Quarter
 - Instructor of Pre-Major in Astronomy Program (ASTR 192) during Fall Quarter
- Universidad Nacioinal Autonoma de Mexico**, Research Assistant Ciudad de Mexico, Mexico
June 2024 – Sept 2024
- Instituto de Ciencias Nucleares; Advisor: Antigona Segura
 - Used Python and Fortran to understand how energetic particles from solar flares could affect atmospheric nitrogen on exoplanets.
- Stanford University**, Research Assistant Palo Alto, CA, USA
June 2023 – Sept 2023
- Earth and Planetary Sciences; Advisor: Laura Schaefer
 - Used MATLAB to add the physics of CO2 outgassing from planetary interior on a magma ocean code.
- University of Washington**, Teaching Assistant Seattle, WA, USA
Sept 2019 – May 2021
- Taught the Activity portion of Introduction to Astronomy for undergraduate students in-person and online.
 - Moderated discussions for 30+ students in the online course.
 - Migrated Introduction to Astrobiology course from in-person to online Canvas platform during the COVID pandemic and the course is still available online.
- CERN**, Research Intern Geneva, Switzerland
June 2019 – Sept 2019
- ATLAS group; Advisor: Matt LeBlanc
 - Used C++ to process and clean data from particle collisions gathered by the ATLAS detector.
- NASA Jet Propulsion Laboratory**, Visiting Research Student La Canada, CA, USA
Aug 2018 – May 2019
- Astrobiology Division; Advisor: Renyu Hu
 - Used MATLAB to calculate the effect of lightning on atmospheric nitrogen when bodies of water are present on the exoplanet around GJ 876.

California State University Los Angeles , Research Assistant • Physics Department; Advisor: Luca Ricci • Used ALMA radio telescope data to study the chemical components of protoplanetary disks.	Los Angeles, CA Jan 2018 – June 2018
Universidad de Puerto Rico en Cayey , Student Researcher • Laboratorio de Modelaje y Simulaciones Teoricas • Modeled rate of reaction for different configurations of atmospheric CO₂ and H₂O.	Cayey, PR, USA Aug 2015 – May 2017
NASA Goddard Space Flight Center , Summer Research Intern • Planetary Systems Laboratory; Advisor: Paul Romani • Used data from the Cassini-Huygens mission to constrain the source of stratospheric water of Saturn	Greenbelt, MD, USA June 2016 – Aug 2016
The Washington Center , International Relation Interns • Social Inclusion Division at Organization of American States; Advisor: Alexandra Barrantes • Assisted in communications between the Division of Social Inclusion and Diplomats of the American States for diplomatic meetings. • Assisted in Translating, proofreading, and formatting the book "Protocolo San Salvador" • Live English scribe for the Secretary of Social Inclusion Meetings for the American States.	Washington, DC, USA Jan 2015 – June 2015
San Jose State University , Summer Research Intern • Laboratory of Neuroscience; Advisor: Katherine Wilkinson • Used mice leg muscles to study the effects of lactic acid on nociceptors III and IV neurons and how obesity could be related to associated muscular and joint pain.	San Jose, CA, USA June 2014 – Aug 2014

Publications

Climates of Gl 514 b (in-prep)

Hector Delgado Diaz, Rory Barnes, Russell Deitrick, Mario Damasso

Stability of Nitrogen in Planetary Atmospheres in Contact with Liquid Water

Nov 2019

Renyu Hu, **Hector Delgado Diaz**

[10.3847/1538-4357/ab4cea](https://doi.org/10.3847/1538-4357/ab4cea) [↗](#) (The Astrophysical Journal)

Awards and Fellowships

Beca Josep Comas i Sola Escuela de Astrobiologia	Santander, Spain July 2023
Bonderman Fellowship Award	Apr 2021
• Selected out of 200 applicants to travel the world for 8 months as part of a cultural experience. • Dates traveled: September 2023 - May 2024.	
Department of Astronomy Teacher of the Year Award	Seattle, WA, USA Sept 2020
ARCS Foundation Fellowship	Seattle, WA, USA Aug 2019
• Selected for a 3-year fellowship totaling \$17,500.	
GEM Fellow	Seattle, WA, USA Aug 2019
Exo-Atmospheres Summer School Scholarship	Chur, Switzerland June 2019
Hispanic Scholarship Fund Award	Los Angeles, CA, USA

Aug 2018

Talks and Posters

Climates of Gl 514 b American Astronomical Society 245th Winter Meeting, Seattle, WA, USA	Jan 2025
Climates of Gl 514 b NASA Goddard: ROCKE-3D Journal Club	Aug 2024
Climates of Gl 514 b Simposio en Materiales y Fisicoquimica - Sociedad Dominicana de Fisica	Aug 2024
FLUXS - Gas Fluxes for Understanding and Experimenting the Subsurface Mars Ideation Factory, NASA Goddard	July 2023
Climates of Gl 514 b Qualifying Exam, University of Washington	June 2023
Climates of Gl 514 b AbGradCon, San Diego, CA, USA	May 2023
Climates og Gl 514 b (poster) American Astronomical Society 241st Winter Meeting, Seattle, WA, USA	Jan 2023
Procurando Exoplanetas Habitables con el Telescopio JWST Spanish talk for the Museum of Flight, Seattle, WA, USA	Dec 2021
Searching for Habitable Exoplanets Using the Telescope JWST Museum of Flight, Seattle, WA, USA	Oct 2021
Characterizing Exoplanetary Atmospheres with JWST Town Hall Seattle, WA, USA	May 2021
Caracterizando Atmosferas de Exoplanetas con JWST Sociedad Dominicana de Fisica, presentacion en linea	Jan 2021
Stability of Nitrogen in Planetary Atmospheres in Contact with Liquid Water American Astronomical Society 233rd Winter Meeting, Seattle, WA, USA	Jan 2019
Stability of Nitrogen in Planetary Atmospheres in Contact with Liquid Water American Physical Society Bridge Program Conference, Mountain View, CA, USA	Nov 2018
Saturn's Stratospheric Oxygen Compounds Division of Planetary Science, Pasadena, CA, USA	Oct 2016
Effect of Obesity and Lactic Acid on Muscle's Nociceptors III and IV neurons Annual Biomedical Research Conference for Minority Students, San Antonio, TX, USA	Nov 2015
Effect of Obesity and Lactic Acid on Muscle's Nociceptors III and IV Society for the Advancement of Chicanos and Native Americans in Science, Los Angeles, CA, USA	Oct 2015

Outreach and Volunteering Experience

Peer Reviewer, Astrophysical Journal Supplement Series Invited to review a manuscript on distance estimates for RR Lyrae Stars	Feb 2025
Origins: Life in the Universe - Co-founder of the Astrobiology Registered Student Organization at the UW - Graduate Student Advisor position	Dec 2024 – present
UW Astronomy Graduate Senator Member of the Diversity Comittee and Travel Grants Comittee	Sept 2024 – present

- Monthly meetings held for the General Senate and weekly meetings for each committee
- Co-writer of approved resolution

NASA Subject Matter Expert Volunteering

Oct 2021 – present

VPLanet Workshop - Tech Support

Aug 2023

UW Astronomy Graduate Senator

Sept 2020 – May 2021

Professional Development

Mars Ideation Factory

Greenbelt, MD, USA
Aug 2023

Astrobiology Summer School

Santander, Spain
July 2023

Carl Sagan Summer Workshop

Pasadena, CA, USA
July 2023

Exo-Atmospheres Summer School

Chur, Switzerland
June 2019

Research Mentorship

Nathaniel Brown (University of Washington)

2025

Kyler Reynolds (Everett College)

2024-2025

Work Experience

Fusion Academy, High School Science Teacher

Pasadena, CA, USA
June 2022 – Sept 2023

- Taught high school students Science courses: Astronomy, Chemistry, Physics, Biology, Math (Algebra-Calculus), and Earth and Space Science.
- Served as a substitute teacher for Spanish and Economy.

Griffith Observatory, Museum Guide and Online Support

Los Angeles, CA, USA
June 2021 – Sept 2023

- Worked explaining astronomy history, instruments, and science to the general public in English and Spanish.
- Worked as online support for the Griffith Museum Online School Program.

References

- **Dr. Rory Barnes** (UW) - Ph.D. Advisor: rkb9@uw.edu 
- **Dr. Jessica Werk** (UW) - Astronomy Chair: jwerk@uw.edu 
- **Dr. Eric Agol** (UW) - Graduate Advisor: agol@uw.edu 